

User Manual for

Fast3VmrMLM

Fast three-Variance-components **m**ulti-locus
random-SNP-effect **M**ixed **L**inear **M**odel tools for
genome-wide association study
(**version 3.0**)

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Disclaimer: The software has undergone comprehensive testing by Yuan-Ming Zhang's Lab at College of Plant Science and Technology, Huazhong Agricultural University. The results obtained from the software are generally reliable, correct, and appropriate. However, it is important to note that these results are not guaranteed for specific datasets. We strongly recommend that users integrate Fast3VmrMLM results with those from other software packages, i.e., 3VmrMLM and q3VmrMLM.

Download website:

<https://github.com/YuanmingZhang65/Fast3VmrMLM>

Citation:

Wang JT[#], Chen Y[#], Shu G[#], Zhao M[#], Zheng A, Chang X, Li G, Wang Y^{*} and Zhang YM^{*}. Fast3VmrMLM: A fast algorithm that integrates genome-wide scanning with machine learning to accelerate gene mining and breeding by design for polygenic traits in large-scale GWAS datasets. **Plant Communications** 2025; 6: 101385.

Wang JT[#], Han XL[#], Zhao MM, Zhang HQ, Chen Y, Jiang QY and Zhang YM^{*}. A fast method for breeding by design via G×E interactions detected in large-scale climatic, phenomic and genomic data. **National Science Review** 2026, 13(10), nwag095.

Jing-Tian Wang[#], Miao-Miao Zhao[#], Xue-Lian Han[#], Xiao-Yu Chang, Xiao-Shuang Ma, Qiu-Yun Jiang, Yuan-Ming Zhang^{*}. Fast3VmrMLM enables statistically powerful and computationally efficient large-scale marker-pair epistasis GWAS in sizable populations. **Plant Communications**, 2026, in press.

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1 Introduction

1.1 Why Fast3VmrMLM?

Fast3VmrMLM (**Fast three-V**ariance-component multi-locus random-SNP-effect Mixed Linear Model) is an R package designed for identifying QTNs in a single dataset (one environmental dataset or the BLUP / mean data from multiple environmental datasets), QTN-by-environment interactions (QEIs) in multi-environment joint analysis, and QTN-by-QTN interactions (QQI) via fast, efficient, and big-data genome-wide association studies (GWAS).

This method considers all possible genetic effects and controls for all possible polygenic backgrounds. These effects are replaced by an effect vector, while their polygenic backgrounds are replaced by a vector-based polygenic background vector. Thus, 5, 10, and 15 variance components, in QTN, QEI, and QQI detection, respectively, are compressed into 3 variance components. This is named a compressed (three) variance component mixed linear model.

Fast3VmrMLM v3.0 with four modules is used to identify **QTNs, QEIs, and QQIs** via SNPs (Fast3VmrMLM), bin/gene haplotypes (Fast3VmrMLM-Hap), pedigree-inferred marker genotypes in crop breeding (Fast3VmrMLM-NCII), and other molecular markers (Fast3VmrMLM-mQTL):

1) Fast3VmrMLM: The genome-wide scanning plus machine learning framework was integrated with advanced computational and statistical techniques to identify QTNs, QEIs, and QQIs for complex traits using a fast, efficient, and big-data GWAS algorithm, Fast3VmrMLM.

2) Fast3VmrMLM-Hap: The Fast3VmrMLM-Hap module is used to associate the haplotypes of bins constructed by adjacent linkage disequilibrium markers with the trait of interest.

3) Fast3VmrMLM-NCII: The Fast3VmrMLM-NCII module is used to associate the marker genotypes inferred from parental marker genotypes with the trait of interest in the North Carolina Design II scheme or other mating designs (crop breeding populations).

4) Fast3VmrMLM-mQTL: Fast3VmrMLM-mQTL extends the analysis to functional haplotypes/genotypes, thereby capturing associations that may be missed by conventional haplotype-based methods. Specifically, when multiallelic variants, k -mers, regulatory elements, genes, and polyploid genotypes/haplotypes are encoded as $G_1, G_2, \dots, G_k$, Fast3VmrMLM-mQTL

can be used to identify their associations with quantitative/omics traits. Once the molecular QTL genotypes were integrated into a *.VCF file format, Fast3VmrMLM-mQTL can handle the multi-type of them for QTN, QEI, and QQI detection. Fast3VmrMLM-mQTL also support PGEN/PVAR/PSAM input for gene functional haplotype-based GWAS, see § 2.2.

All of above three modules support **single-environment GWAS** (identifying QTNs and QTN-by-meteorological factors), **multi-environment joint GWAS** (identifying QTNs and QEIs), and **epistatic GWAS** (identifying QTNs and QQIs).

Fast3VmrMLM v3.0 works on the **Windows** and **Linux** systems.

1.2 Getting started

Fast3VmrMLM v3.0 (<https://github.com/YuanmingZhang65/Fast3VmrMLM>) is a package that runs in the R environment on the Linux/Windows system, which can be freely downloaded from the above website or requested from the maintainer, Dr. Yuan-Ming Zhang at College of Plane Science and Technology, Huazhong Agricultural University (450625680@qq.com; soy Zhang@mail.hzau.edu.cn).

1.2.1 Installation in the Linux

Within the Linux system, the Fast3VmrMLM software can be installed by the following four steps:

1) To download and install **Miniconda3** at <https://repo.anaconda.com/miniconda/>, in which the recommended installer file is [Miniconda3-py38_23.11.0-2-Linux-x86_64](#). Detailed for installing Miniconda3 were shown in §5.1.

2) To create a new **conda environment** named Fast3VmrMLM and install **R version 4.3** by the following code:

```
conda create -n "Fast3VmrMLM" r-essentials r-base=4.3  
conda activate Fast3VmrMLM
```

3) To install the dependency packages.

- Install dependency packages by following codes in Linux environment (or by *install.packages* in R environment):

```
conda install -c conda-forge r-Rcpp=1.1.0 r-RcppArmadillo=15.0.2_1  
r-data.table=1.15.2 r-RcppParallel=5.1.9 r-MASS=7.3_60.0.1 r-openxlsx=4.2.8  
r-BH=1.87.0_1 r-BEDMatrix=2.0.4 r-pgenlibr=0.5.3
```

4) Downloading Fast3VmrMLM from Github and installing it:

- Downloading via R code in R environment (R version 4.3):

```
download.file("https://codeload.github.com/YuanmingZhang65/Fast3VmrMLM/zip/refs/heads/main",destfile="Fast3VmrMLM-main.zip",mode="wb",method="libcurl")
```
- Installing Fast3VmrMLM in R environment (R version 4.3):

```
unzip("Fast3VmrMLM-main.zip")  
unzip("Fast3VmrMLM-main/Fast3VmrMLM_Linux_3.0.zip")  
install.packages("Fast3VmrMLM", repos = NULL)
```

Users can install and unzip ARM architecture Fast3VmrMLM in the R environment (R version 4.3) if the server uses an ARM architecture CPU:

```
unzip("Fast3VmrMLM-main/Fast3VmrMLM_Linux_arm_3.0.zip")  
install.packages("Fast3VmrMLM", repos = NULL)
```

1.2.2 Installation in the **Windows** system

1) Download the installer of R 4.3.2 at <https://cran.r-project.org/bin/windows/base/old/4.3.2/R-4.3.2-win.exe>, and install R 4.3.2.

2) Download the installer of Rtools 4.3 at <https://cran.r-project.org/bin/windows/Rtools/rtools43/files/rtools43-5976-5975.exe>, and install R 4.3.2.

3) To install the dependency packages in R environment (R version 4.3.2):

- ```
pkgs=c("Rcpp_1.0.14","RcppArmadillo_14.0.2-1","RcppParallel_5.1.10","data.table_1.17.0","MASS_7.3-60.0.1","openxlsx_4.2.8","BH_1.87.0-1","stringi_1.8.7","zip_2.3.2","BEDMatrix_2.0.4","crochet_2.3.0","pgenlibr_0.4.0")
```
- ```
mirror="https://mirrors.tuna.tsinghua.edu.cn/CRAN/bin/windows/contrib/4.3/"
```
- ```
install.packages(paste0(mirror, pkgs, ".zip"), repos=NULL, type="win.binary")
```

4) Downloading Fast3VmrMLM from Github and installing it:

- Downloading via R code in R environment (R version 4.3):  

```
download.file("https://codeload.github.com/YuanmingZhang65/Fast3VmrMLM/zip/refs/heads/main",destfile="Fast3VmrMLM-main.zip",mode="wb",method="libcurl")
```
- Installing Fast3VmrMLM in R environment (R version 4.3):

```
unzip("Fast3VmrMLM-main.zip")
install.packages("Fast3VmrMLM-main/Fast3VmrMLM_3.0.zip",repos=NU
LL,type="win.binary")
```

Before performing epistasis GWAS, users must install PLINK v1.9, which is available at <https://www.cog-genomics.org/plink/1.9/>.

If you have any questions about installation or anything else, please don't hesitate to contact Prof. Yuan-Ming Zhang ([soy Zhang@mail.hzau.edu.cn](mailto:soy Zhang@mail.hzau.edu.cn); [450625680@qq.com](mailto:450625680@qq.com)) or Dr. Jing-Tian Wang ([1390032282@qq.com](mailto:1390032282@qq.com)).

**User Manual file** Users can decompress the Fast3VmrMLM package and find the User Manual file (name: Instruction 3.0.pdf) in the folder of ".../Fast3VmrMLM/inst". Users also download it from the Fast3VmrMLM website.

**Example datasets** Users can decompress the Fast3VmrMLM package and find the example datasets in the folder of ".../Fast3VmrMLM/extdata".

### 1.2.3 Run Fast3VmrMLM

Once the software Fast3VmrMLM is installed, users may run it using two commands:

```
library(Fast3VmrMLM)
```

```
Fast3VmrMLM(***) (please see the example of § 2.2)
```

Before using Fast3VmrMLM in Linux, make sure to activate the conda environment of Fast3VmrMLM: [conda activate Fast3VmrMLM](#). Users need to run [library\(Fast3VmrMLM\)](#) every time before utilizing the Fast3VmrMLM software package.

## 2 Function Fast3VmrMLM

Three main functions are provided: *Fast3VmrMLM()* for single-environment GWAS, *Fast3VmrMLM\_MEJA()* for multi-environment joint GWAS and *Fast3VmrMLM\_QQI()* for epistasis GWAS. They share largely similar parameters; the parameters specific to *Fast3VmrMLM\_MEJA()* and *Fast3VmrMLM\_QQI()* are explicitly indicated in the table below.

### 2.1 Parameter settings

| Parameter           | Meaning                                                                                                                                                                                                                                                                                                                                             | File format                                                                                                                                                                                                         | Note              |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>fileGen</b>      | Name & path of genotypic file in your device, e.g., <i>fileGen="D:/Users/Genotype"</i> . <b>All file paths should exclude "~".</b>                                                                                                                                                                                                                  | PLINK binary files:<br><b>Genotype.bed+Genotype.bim+Genotype.fam</b>                                                                                                                                                |                   |
| <b>filePhe</b>      | Name & path of trait phenotypic file in your device, e.g., <i>filePhe="D:/Users/Phenotype.csv"</i> .                                                                                                                                                                                                                                                | *.csv (Phenotypic values. Row: individuals; Column: traits)                                                                                                                                                         | <b>Table 1</b>    |
| <b>fileKin</b>      | Name & path of individual kinship file in your device, e.g., <i>fileGRM="D:/Users/GRM.csv"</i> or <i>fileGRM=NULL</i> .                                                                                                                                                                                                                             | *.csv (GRM. Row & Column: individuals)                                                                                                                                                                              | <b>Table 2</b>    |
| <b>filePS</b>       | Name & path of covariates file in your device, e.g., <i>filePS="D:/Users/covariates.csv"</i> or <i>filePS=NULL</i> .                                                                                                                                                                                                                                | *.csv (Population structure, or principal components, or evolutionary population structure. Row: individuals; Column: sub-populations / principal components / evolutionary types or covariates, e.g., sex and age) | <b>Tables 3-5</b> |
| <b>filePedigree</b> | <i>filePedigree="D:/Users/Pedigree.csv"</i> when using the NC II genetic mating design population, and <i>filePedigree=NULL</i> (the default value) for all others.                                                                                                                                                                                 | *.csv (Pedigree of NCII population consists of 3 individual-ID columns: the female & male parents, and their F <sub>1</sub> offspring)                                                                              | <b>Table 7</b>    |
| <b>PopStrType</b>   | The types of population structure include "PopStrType=Q" ( <i>Q matrix or ancestry coefficient matrix calculated by Admixture software</i> ), "PopStrType=PC" ( <i>principal components or covariates</i> ), and "PopStrType=Evol" ( <i>evolutionary population structure</i> ).                                                                    |                                                                                                                                                                                                                     |                   |
| <b>fileOut</b>      | Save path of the result in your device, e.g., <i>"D:/Users/"</i> .                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                     |                   |
| <b>genoType</b>     | Setting the algorithm as " <b>SNP</b> " (SNP-based Fast3VmrMLM module), " <b>Hap</b> " (Haplotype-based Fast3VmrMLM-Hap module) and " <b>molecular</b> " (molecular-based Fast3VmrMLM-mQTL module)                                                                                                                                                  |                                                                                                                                                                                                                     |                   |
| <b>population</b>   | population="NCII" when using the NCII design, and population="nature" (the default value) for all others                                                                                                                                                                                                                                            |                                                                                                                                                                                                                     |                   |
| <b>trait</b>        | Traits analyzed from number 1 to number 2, e.g., <i>trait=1:3</i> : users analyze the first to third traits.                                                                                                                                                                                                                                        |                                                                                                                                                                                                                     |                   |
| <b>svrad</b>        | A physical distance of sliding window for removing potential candidate variants with the collinearity of the most significant one. Default value is <i>svrad=2.0e+4</i> (bp). Users can obtain more potential associated variants by setting a small value of SearchRadius.                                                                         |                                                                                                                                                                                                                     |                   |
| <b>svpal</b>        | A critical <i>P</i> -value (default <i>svpal=1.0e-2</i> ) to select all potentially associated variants/loci in genome-wide single-variant scanning (the first step). The parameter <i>svpal</i> may be changed based on sample size see §2.3. The <i>P</i> -value differs from 1.0e-5 for large population (>4000) to 1.0e-2 for small population. |                                                                                                                                                                                                                     |                   |

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>svmlod</b>                 | A critical LOD score is set as <b>svmlod=3</b> (default) for identifying suggested variants/loci, while a critical P-value is set as <b>svmlod=0.05/m</b> (Bonferroni correction; m: the number of markers) for identifying significant variants/loci.                                                                                                                                                                                                                              |
| <b>SampleMarkersforGRM</b>    | A parameter indicates whether a subset of variants is used to construct kinship ( <b>TRUE</b> ) or not ( <b>FALSE</b> ). Default value is <b>SampleMarkersforGRM=FALSE</b> .                                                                                                                                                                                                                                                                                                        |
| <b>SampleMarkersforGRMNum</b> | A parameter (>0) indicates the number of variants sampled to calculate kinship matrix. Only used when SampleMarkersforGRM=TRUE. Default value is <b>SampleMarkersforGRMNum=1.0e+4</b> .                                                                                                                                                                                                                                                                                             |
| <b>c_threshold</b>            | A parameter for evaluating linkage disequilibrium of adjacent variants when constructing bin-based haplotypes, ranging from 0 to 1. Default value is <b>c_threshold=0.7</b> .                                                                                                                                                                                                                                                                                                       |
| <b>numofHaplotypes</b>        | A parameter for setting the number of haplotypes for each bin genotype in the Fast3VmrMLM-Hap and Fast3VmrMLM-mQTL module. Default value is <b>numofHaplotypes=3</b> for Fast3VmrMLM-Hap and <b>numofHaplotypes=6</b> for Fast3VmrMLM-mQTL.                                                                                                                                                                                                                                         |
| <b>nThreads</b>               | A parameter indicates the number of cores used in parallel computing. Default value is set as <b>nThreads=20</b> .                                                                                                                                                                                                                                                                                                                                                                  |
| <b>DrawPlot</b>               | A parameter indicates whether drawing and outputting the Manhattan plot based on the GWAS results. Default value is <b>DrawPlot=FALSE</b> .                                                                                                                                                                                                                                                                                                                                         |
| <b>Plotformat</b>             | A parameter indicates the format of the Manhattan plot. Default value is <b>Plotformat=*.tiff</b> .                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>MGinputClass</b>           | Setting the input file format “ <b>bed</b> ” (PLINK binary file) and “ <b>vcf</b> ” (.vcf format file), when <b>variableType=“molecular”</b> in the Fast3VmrMLM-mQTL module. Default value is <b>inputClass=“vcf”</b> .                                                                                                                                                                                                                                                             |
| <b>filegeneRegion</b>         | Setting the input file that indicates the genome intervals of each gene or molecular marker when <b>inputClass=“bed”</b> in the Fast3VmrMLM-mQTL module.                                                                                                                                                                                                                                                                                                                            |
| <b>n_en</b>                   | A parameter for Fast3VmrMLM_MEJA function, a vector setting the environment number of each trait, etc., when there were two traits measured under three and four environments, respectively, setting <b>n_en=c(3,4)</b> .                                                                                                                                                                                                                                                           |
| <b>effect_out</b>             | <b>effect_out=TRUE</b> : Output all the effects of genome-wide scanning into the ‘intermediate results’ file, including additive and dominant effects in single-environment GWAS, and additive, dominant, additive×environment interaction and dominant×environment interaction effects in multi-environment GWAS. <b>effect_out=FALSE</b> : Do not output all the effects of genome-wide scanning into the ‘intermediate results’ file. Default value is <b>effect_out=FALSE</b> . |
| <b>thinMarker</b>             | A parameter indicating whether to produce a pruned subset of markers based on linkage equilibrium before epistasis GWAS. Only for <b>Fast3VmrMLM_QQI()</b> function. Default value is <b>thinMarker=TRUE</b> .                                                                                                                                                                                                                                                                      |
| <b>LDthin_max.dist</b>        | When <b>thinMarker=TRUE</b> , this parameter indicated the physical distance of sliding window used in the pruned subset process. Default value is <b>LDthin_max.dist=200e+3</b> .                                                                                                                                                                                                                                                                                                  |
| <b>LDthin_threshold</b>       | When <b>thinMarker=TRUE</b> , This parameter indicates the correlation coefficient threshold used to delete markers within the sliding window. Default value is <b>LDthin_threshold=0.8</b> .                                                                                                                                                                                                                                                                                       |
| <b>plinkdir</b>               | The absolute path of PLINK v1.9 software. For example, <b>plinkdir=“/home/plink/”</b> .                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>plinkepi1</b>              | The threshold of the PLINK epistasis scanning. Default value is <b>plinkepi1=1e-5</b> .                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>heterozygote_rate</b>      | Only consider the additive effect for the loci with heterozygote rate less than <b>heterozygote_rate</b> . The default value is <b>heterozygote_rate=0.0001</b> .                                                                                                                                                                                                                                                                                                                   |

## 2.2 Running codes

### 2.2.1 Running codes for single-environment GWAS in R v4.3

The running code for Fast3VmrMLM module is as follows:

```
Fast3VmrMLM(fileGen="/home/Genotype",filePhe="/home/phenotype.csv",fileKin=NULL,
filePS="/home/PopStr.csv",PopStrType="PC",fileOut="/home/",genoType="SNP",
,trait=1,svrad=2e+4,svpal=1e-2,svmlod=3,nThreads=20,DrawPlot=FALSE, Plotformat
="*.tiff")
```

To create a Manhattan plot, please set the parameter, *DrawPlot=TRUE*. This will output a Manhattan plot for each trait. Alternatively, users can draw the plot using the *Fast3VmrMLM\_manplot()* function, as detailed in §4.1 of Instruction.

The running code of Fast3VmrMLM when using NCII design is as follows:

```
Fast3VmrMLM(fileGen="/home/Genotype",filePhe="/home/phenotype.csv",fileKin=NULL,
filePS="/home/PopStr.csv",filePedigree="/home/Pedigree.csv",PopStrType="PC",
,population="NCII",fileOut="/home/",genoType="SNP",trait=1,svrad=2e+4,svpal=1e-2
,svmlod=3,nThreads=20,DrawPlot=FALSE,Plotformat="*.tiff")
```

When using **partial NCII breeding population datasets** via the Fast3VmrMLM module, the genotype file (*fileGen*) should have parental genotypes, and offspring's genotypes can be inferred from their parents by Fast3VmrMLM according to the pedigree (*filePedigree*).

The running code for Fast3VmrMLM-Hap module:

```
Fast3VmrMLM(fileGen="/home/Genotype",filePhe="/home/phenotype.csv",fileKin=NULL,
filePS="/home/PopStr.csv",PopStrType="PC",fileOut="/home/",genoType="Hap",
,trait=1,svrad=2e+4,svpal=1e-2,svmlod=3,c_threshold=0.7,numofHaplotypes=3)
```

The code for running the Fast3VmrMLM-mQTL module with VCF input:

```
Fast3VmrMLM_SV(fileGen="home/genenotype_mQTL_vcf.vcf",filePhe="home/trait_
mQTL_vcf.csv",filePS=NULL,fileOut="home/",trait=1:3,numofHaplotypes=6,svpal=0.0
1,svrad=1e+4,svmlod=3)
```

The code for running the Fast3VmrMLM-mQTL module with PGEN/PVAR/PSAM input (for details about this format, see <https://www.cog-genomics.org/plink/2.0/>):

```
Fast3VmrMLM_SV_PGEN(fileGen="genenotype_mQTL_pgen",filePhe="trait_mQTL
```

```
_vcf.csv",fileOut="/home/",trait=1:3,numofHaplotypes=6,svpal=0.01,svrad=1e+4,svmlod=3)
```

Users **must set** "fileGen", "filePhe", "trait", and "fileOut", while the other parameters may be default in function **Fast3VmrMLM** (see § 2.1).

## 2.2.2 Running codes for multi-environment GWAS

The code for **multi-environment GWAS** is the same as that shown above, except for two changes: the function *Fast3VmrMLM()* should be replaced by *Fast3VmrMLM\_MEJA()*, and the *n\_en* parameter should be provided.

The running code for Fast3VmrMLM in multi-environment GWAS is as follows:

```
Fast3VmrMLM_MEJA(fileGen="/home/Genotype",filePhe="/home/phenotype.csv",fileKin=NULL,filePS="/home/PopStr.csv",PopStrType="PC",fileOut="/home/",genoType="SNP",trait=1:2,n_en=c(3,2),svrad=2e+4,svpal=1e-2,svmlod=3,nThreads=20,DrawPlot=FALSE,Plotformat="*.tiff")
```

The running code of Fast3VmrMLM module in multi-environment GWAS when using NCII dataset is as follows:

```
Fast3VmrMLM_MEJA(fileGen="/home/Genotype",filePhe="/home/phenotype.csv",fileKin=NULL,filePS="/home/PopStr.csv",filePedigree="/home/Pedigree.csv",PopStrType="PC",population="NCII",fileOut="/home/",genoType="SNP",trait=1,n_en=c(2),svrad=2e+4,svpal=1e-2,svmlod=3,nThreads=20,DrawPlot=FALSE,Plotformat="*.tiff")
```

The running code for Fast3VmrMLM-Hap module:

```
Fast3VmrMLM_MEJA(fileGen="/home/Genotype",filePhe="/home/phenotype.csv",fileKin=NULL,filePS="/home/PopStr.csv",PopStrType="PC",fileOut="/home/",genoType="Hap",trait=1,n_en=c(2),svrad=2e+4,svpal=1e-2,svmlod=3,c_threshold=0.7,numofHaplotypes=3)
```

The running code for Fast3VmrMLM-mQTL module in multi-environment GWAS with VCF input:

```
Fast3VmrMLM_MEJA_SV(fileGen="/home/Genotype.vcf",filePhe="/home/phenotype.csv",fileKin=NULL,filePS="/home/PopStr.csv",PopStrType="PC",fileOut="/home/",trait=1:2,n_env=c(2, 2),numofHaplotypes=6,svpal=0.01,svrad=2e+4,svmlod=3)
```

The running code for Fast3VmrMLM-mQTL module in multi-environment GWAS with PGEN/PVAR/PSAM input:

*Fast3VmrMLM\_MEJA\_SV\_PGEN*(fileGen="gene\_SV\_pgen",filePhe="quantitative\_SV.csv",fileOut="/home/",trait=1,n\_env=2,numofHaplotypes=6,svpal=0.01,svrad=1e+4,svmlod=3)

### 2.2.3 Running codes for epistasis GWAS

The function *Fast3VmrMLM()* should be replaced by *Fast3VmrMLM\_QQI()*.

The running code for Fast3VmrMLM in epistasis GWAS is as follows:

*Fast3VmrMLM\_QQI*(fileGen="SNPsubset2",filePhe="Sim.csv",filePS="2pca.csv",fileOut="result/",genoType="SNP",trait=1:2,thinMarker=TRUE,LDthin\_threshold=0.8,LDthin\_max.dist=200e3,plinkdir="/home/WJT/plink/",plinkepi1=1e-5,svpal=1e-8,svpal\_ad=1e-3,svrad=1e+4,svmlod=3,svmlod\_QQI=3)

The running code of Fast3VmrMLM-NCII module in epistasis GWAS when using NCII dataset is as follows:

*Fast3VmrMLM\_QQI*(fileGen="sample\_1000",filePhe="Sim.csv",filePS="pca.csv",population="NCII",filePedigree="pedigree.csv",fileOut="/home/",genoType="SNP",trait=1:2,thinMarker=TRUE,LDthin\_threshold=0.8,LDthin\_max.dist=200e3,plinkdir="/home/WJT/plink/",plinkepi1=1e-5,svpal=1e-8,svpal\_ad=1e-3,svrad=10,svmlod=3,svmlod\_QQI=3)

The running code for Fast3VmrMLM-Hap module:

*Fast3VmrMLM\_QQI*(fileGen="SNPsubset2",filePhe="Sim.csv",filePS="pca.csv",fileOut="/home/",genoType="Hap",c\_threshold=0.7,numofHaplotypes=3,trait=1:2,thinMarker=TRUE,LDthin\_threshold=0.8,LDthin\_max.dist=200e3,plinkdir="/home/WJT/plink/",plinkepi1=1e-5,svpal=1e-5,svpal\_ad=1e-3,svrad=10,svmlod=3,svmlod\_QQI=3)

The running code for Fast3VmrMLM-mQTL module in epistasis GWAS with VCF input:

*Fast3VmrMLM\_SV\_QQI*(fileGen="sub300.vcf",filePhe="Sim.csv",filePS=NULL,fileOut="/home/",trait=1:2,plinkdir="/home/WJT/plink/",plinkepi1=1e-5,svpal=1e-5,svrad=1e+4,svmlod=3,numofHaplotypes=6)

The running code for Fast3VmrMLM-mQTL module in epistasis GWAS with PGEN/PVAR/PSAM input:

*Fast3VmrMLM\_SV\_QQI\_PGEN*(fileGen="sv.lumpy2.sub300.keep\_QTN\_QQI\_pgen",filePhe="/quantitative\_Simulation\_SV\_QQI\_1.csv",fileOut="/home/",plinkdir="/home/plink/",trait=1,threads=2,svpal=1e-5,svrad=1e+4,svmlod=3,numofHaplotypes=6)

## 2.3 Parameter adjustment

When applying the new method, three parameters may be adjusted, including **svpal**, **c\_threshold** and **numofHaplotypes**. First, the **svpal** was used to set the P-value threshold for potential QTNs/QEIs in the first step. The recommended threshold depends on the population size: the larger the population size, the smaller the threshold. For example, **svpal = 0.05 or 0.01 for a population size of less than 4,000**, and **svpal = 1e-5 for a population size of more than 20,000**. Second, the **c\_threshold** and **numOfHaplotypes** parameters can be adjusted when constructing haplotypes in Fast3VmrMLM-Hap. **c\_threshold** (0 to 1; 0.7 is recommended) is used to determine bin markers: the smaller the value, the larger the number of bin markers. The **numofHaplotypes** parameter determines the maximum number of haplotypes (e.g., three). We suggest to set **svpal = 5e-8** in epistasis GWAS.

When Fast3VmrMLM-mQTL is used for epistasis GWAS, **svpal** can be set to **1e-5** and **svrad** can be set to **1e+5** for population sizes below 4,000 and to **0.05/[ $m(m-1)/2$ ]** and **svrad** can be set to **1e+4** for population sizes above 4,000.

## 2.4 Data input format

### Format for genotypic dataset “fileGen”

The file type of genotypes is "plink binary format" (Genotype.bed + Genotype.bim + Genotype.fam), [which can be found the introduction from PLINK v1.9 \(https://www.cog-genomics.org/plink2/\)](https://www.cog-genomics.org/plink2/).

### Format for genotypic dataset “filePhe” (Table 1.1 and Table 2.2)

The file type of phenotypes for complex trait is \*.csv, following the format outlined in Table 1.1. The first row in the first column: "<Phenotype>"; the second to  $n$ th rows in the first column: individual IDs or names, such as Ind46. The first row in other columns: trait names, such as “trait1”, and the second to  $n$ th rows in other columns: values of phenotypic traits. The phenotypes missed: “NA”. The phenotype file format for **multi-environment joint GWAS** is shown in Table 1.2, where each column represents one trait measured under one environment. For example, Table 1.2 includes two traits (trait1 and trait2)

measured across two and three environments, respectively, and the `n_en` parameter in *Fast3VmrMLM\_MEJA()* function should be set as: `n_en=c(2,3)`

**Table 1.1. The format of phenotypic dataset**

| <Phenotype> | trait1 | trait2 | trait3 | ... |
|-------------|--------|--------|--------|-----|
| Ind46       | 91.03  | 88.32  | 87.67  | ... |
| Ind52       | 103.11 | 103.10 | 98.36  | ... |
| Ind57       | 92.07  | 116.30 | NA     | ... |
| Ind64       | 128.5  | 101.20 | 101.02 | ... |
| Ind68       | 95.84  | 91.74  | 94.50  | ... |
| ⋮           | ⋮      | ⋮      | ⋮      | ... |

**Table 1.2. The format of phenotypic dataset for multi-environment GWAS**

| <Phenotype> | trait1_env1 | trait1_env2 | trait2_env1 | trait2_env3 | trait2_env3 | ... |
|-------------|-------------|-------------|-------------|-------------|-------------|-----|
| Ind46       | 91.03       | 88.32       | 116.30      | NA          | 87.67       | ... |
| Ind52       | 103.11      | 103.10      | 101.20      | 103.11      | 98.36       | ... |
| Ind57       | 92.07       | 116.30      | 92.07       | 87.67       | NA          | ... |
| Ind64       | 128.5       | 101.20      | 128.5       | 103.10      | 101.02      | ... |
| ⋮           | ⋮           | ⋮           | ⋮           | ⋮           | ⋮           | ... |

### Format for genotypic dataset “fileKin” (Table 2)

The “fileKin” should be a file with \*.csv format. All the kinship coefficients are listed as an  $n \times n$  matrix. Both rows and columns represent individuals that arranged in the order of the \*.fam file. The parameter “fileKin=NULL” indicates that the kinship matrix is calculated by the “Fast3VmrMLM” software. When `fileKin="D:/Users/kinship.csv"`, the kinship matrix with name kinship.csv is uploaded from the folder "D:/Users". If the number and order of individuals in the "kinship.csv" file do not match those in the phenotypic files, our software will match them.

**Table 2. The format of knship dataset**

| Ind46       | Ind52       | Ind12       | Ind143      | Ind57       | Ind67 |
|-------------|-------------|-------------|-------------|-------------|-------|
| 1           | 0.700361011 | 0.599277978 | 0.675090253 | 0.620938628 | ...   |
| 0.700361011 | 1           | 0.620938628 | 0.666064982 | 0.653429603 | ...   |
| 0.599277978 | 0.620938628 | 1           | 0.561371841 | 0.5433213   | ...   |
| 0.675090253 | 0.666064982 | 0.561371841 | 1           | 0.615523466 | ...   |
| 0.620938628 | 0.653429603 | 0.5433213   | 0.615523466 | 1           | ...   |
| ⋮           | ⋮           | ⋮           | ⋮           | ⋮           | ...   |

### Q matrix format for dataset “filePS” (Table 3)

The Q matrix dataset in Table 3 consists of a  $(n+1) \times (k+1)$  matrix, where  $n$  is sample size (the number of the above common individuals), and  $k$  is the number of sub-populations. The first column is “<Structure>” and individual IDs or names. In the 2nd to  $(k+1)$ -th columns, “Q1” to “Qk” indicate sub-populations. In the second row, “0.014”, “0.972” and “0.014” are posterior probabilities that the individual “33-16” is belong to the 1st, 2nd, and 3rd subpopulations, respectively.

**Table 3. The Q matrix format of dataset filePS**

| <Structure> | Q1    | Q2    | Q3    |
|-------------|-------|-------|-------|
| Ind46       | 0.014 | 0.972 | 0.014 |
| Ind52       | 0.003 | 0.993 | 0.004 |
| Ind57       | 0.071 | 0.917 | 0.012 |
| Ind64       | 0.035 | 0.854 | 0.111 |
| ⋮           | ⋮     | ⋮     | ⋮     |

### Principal components or covariates for dataset “filePS” (Table 4)

The principal components or covariates dataset in Table 4 consists of a  $(n+1) \times k$  matrix, where  $n$  is sample size (the number of the above common individuals), and  $k$  is the number of principal components and/or covariates. The first column is “<PCA>” and individual IDs or names. The 2nd to  $k$ -th columns indicate principal components and/or covariates.

**Table 4. The principal components or covariates format of dataset filePS**

| <PCA> | PC1    | PC2    | PC3    | ... |
|-------|--------|--------|--------|-----|
| Ind46 | 0.306  | 0.029  | 0.226  | ... |
| Ind52 | -0.708 | -0.271 | 1.413  | ... |
| Ind57 | -2.330 | 0.116  | -0.824 | ... |
| Ind64 | 1.059  | 0.470  | -0.135 | ... |
| ⋮     | ⋮      | ⋮      | ⋮      |     |

### The evolutionary population structure for dataset “filePS” (Table 5)

The evolutionary population structure dataset in Table 5 consists of a  $(n+1) \times 2$  matrix, where  $n$  is sample size and the number of categories for variables is the number of sub-populations. The first column is “<EvolPopStr>” and individual IDs or names. The 2nd column indicates the evolutionary

sub-population. Other population structure described by character type of variables are supported in this format.

**Table 5. The evolutionary population structure format of dataset filePS**

| <EvolPopStr> | EvolType |
|--------------|----------|
| Ind46        | A        |
| Ind52        | B        |
| Ind57        | A        |
| Ind64        | C        |
| ⋮            | ⋮        |

**The format of the chromosome intervals of each molecular marker for the parameter filegeneRegion in § 2.1 (Table 6)**

When genoType="molecular" and MGinputClass="bed", the genome intervals of each molecular marker should be set by users via the file in Table 6. The first column is its ID, the second column is its chromosome, the third and the fourth columns are the left and right physical positions, respectively.

**Table 6. The chromosome intervals of each molecular marker**

|                 |   |       |       |
|-----------------|---|-------|-------|
| Zm00001eb000010 | 1 | 34617 | 40204 |
| Zm00001eb000020 | 1 | 41214 | 46762 |
| ⋮               | ⋮ | ⋮     | ⋮     |

**The format of the pedigree file in § 2.1 (Table 7)**

When population="NCII", the filePedigree should be set according to Table 7. It contains three individual ID columns: the female parent (first column), the male parent (second column), and their F1 offspring (third column).

**Table 7. The pedigree file**

|      |       |            |
|------|-------|------------|
| LT16 | LT131 | LT16_LT131 |
| LT07 | LT131 | LT07_LT131 |
| ⋮    | ⋮     | ⋮          |

## 2.5 Result

The results include three files: \*\_intermediate.csv (intermediate results), \*\_result.csv (final results) and a Manhattan plot file.

[\\*\\_intermediate.csv](#): This file contains the results of genome-wide single-variants scanning in the first step. In this file, all the columns are named as “MarkerID” (variants name), “CHR” (Chromosome), “POS” (variants position (bp) on the genome), and “pval” (the  $P$ -value for main-effect variants).

| MarkerID   | CHR | POS     | pval        |
|------------|-----|---------|-------------|
| PZB00859.1 | 1   | 157104  | 0.292043111 |
| PZA01271.1 | 1   | 1947984 | 0.185246808 |
| PZA03613.2 | 1   | 2914066 | 0.99208603  |
| PZA03613.1 | 1   | 2914171 | 0.999987108 |
| ⋮          | ⋮   | ⋮       | ⋮           |

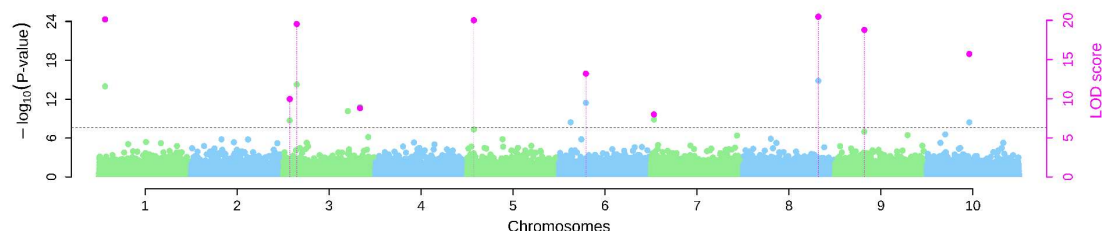
[\\*\\_result.xlsx](#): The final result file for the Fast3VmrMLM module is as follows. All the columns are named as “Marker” (marker name); “Chromosome”; “Position (bp)” (markers position (bp) on the genome); “add” (additive effect); “dom” (dominance effect); “LOD” (LOD score); variance (the variance of each variants); “ $r^2(\%)$ ” (the proportion of total liability variance explained by each variants); “ $P$ -value” (calculated from LOD score using  $\chi^2$  distribution); “significance” (significant (SIG) variants are based on Bonferroni correction, that is, critical  $P$ -value is  $0.05/m$ , where  $m$  is the number of tests or variants, while suggested (SUG) variants are based on  $\text{LOD} \geq 3.0$ , default). “A allele” is the nucleotide base with a dummy variable of 1.0.

| Trait ID | Trait name | Marker     | Chromosome | Position (bp) | LOD     | add     | dom    | variance | $r^2(\%)$ | P-value  | significance | A allele |
|----------|------------|------------|------------|---------------|---------|---------|--------|----------|-----------|----------|--------------|----------|
| 1        | phe_1      | null_653   | 1          | 654           | 31.5242 | -0.4295 | 5.6711 | 7.9444   | 5.0209    | 3.00e-32 | SIG          | C        |
| 1        | phe_1      | null_20728 | 3          | 20729         | 9.9457  | 1.6284  | 2.7827 | 4.3464   | 2.7469    | 1.13e-10 | SIG          | G        |
| 1        | phe_1      | null_52928 | 6          | 52929         | 13.1912 | 3.5793  | -0.163 | 6.4797   | 4.0952    | 6.45e-14 | SIG          | A        |
| 1        | phe_1      | null_78191 | 8          | 78192         | 65.6504 | 6.5965  | -1.579 | 5.054    | 3.1942    | 2.25e-66 | SIG          | T        |

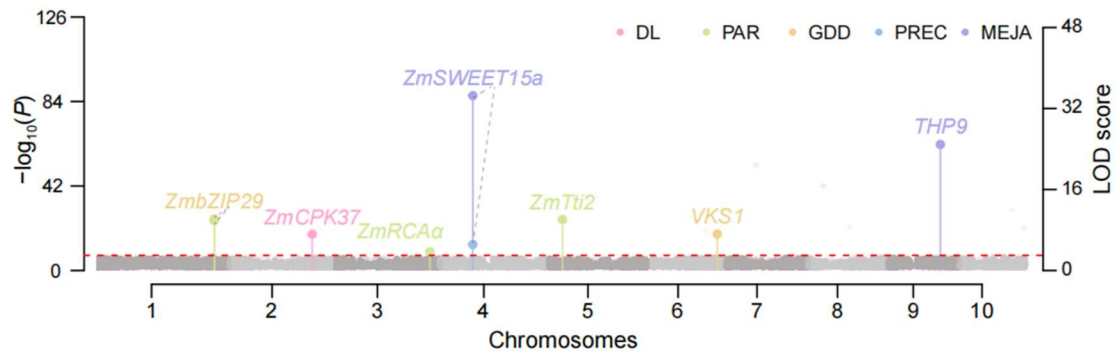
The final result file for the Fast3VmrMLM-Hap module is as follows. The “Marker\_L” and “Marker\_R” mean the markers on the left and right boundaries, respectively, of chromosome region (bin) determined by linkage disequilibrium, while “Position\_L (bp)” and “Position\_R (bp)” are for the left and right genomic positions, respectively. “Hap\_1” to “Hap\_numHap” are the haplotypic effects of the 1st to numHap-th haplotypes, where “numHap” is No. of haplotypes.

| Trait ID | Trait name | Marker_L   | Marker_R  | Chromosome | Position_L (bp) | Position_R (bp) | LOD    | Hap_1  | Hap_2  | Hap_3  | numHap | variance | r <sup>2</sup> (%) | P-value   | significance |
|----------|------------|------------|-----------|------------|-----------------|-----------------|--------|--------|--------|--------|--------|----------|--------------------|-----------|--------------|
| 1        | phe_1      | rs16857692 | rs4668204 | 2          | 170672831       | 170705429       | 5.78   | 0.018  | -0.870 | 0.018  | 3      | 0.274    | 0.1268             | 1.63e-06  | SUG          |
| 1        | phe_1      | rs10439267 | rs6435173 | 2          | 203631692       | 203721357       | 125.53 | -2.213 | -2.595 | -2.213 | 3      | 12.674   | 5.9419             | 2.99e-126 | SIG          |
| 1        | phe_1      | rs1978865  | rs7651321 | 3          | 61127327        | 61256881        | 82.85  | -1.924 | -1.616 | -1.924 | 3      | 6.88     | 3.2254             | 1.42e-83  | SIG          |

\* **\_Manhattan plot**: Y-axis on the left-side reports  $-\log_{10} P$ -values of variants, which are obtained from single-variant genome-wide scanning for all the variants in the first step of Fast3VmrMLM, while Y-axis on the right-side reports LOD scores, which are obtained from likelihood ratio test for suggested and significant variants, with the suggested threshold of LOD = 3.0 (dashed line), in the second step of Fast3VmrMLM. Users can set different LOD threshold by setting **svmlod** (see § 2.1). These LOD scores are shown in points with straight lines. If LOD score  $\geq 20$ , the LOD scores obtained are transformed as  $\text{LOD}' = 20 + (\text{LOD} - 20)/100$  in order that the Manhattan plot is more beautiful.



We recommend marking all the all the significant and suggested QTNs with known and candidate genes in the Manhattan plot and listing all the significantly and suggested associated markers with the traits of interest in a supplemental table. For example, the Manhattan plot below depicts all the known and candidate gene-by-environment interactions, along with their corresponding QEIs/QMIs, as shown in **Figure S2 of Wang et al. (2026)**.



### 3 References

- [1] Wang JT<sup>#</sup>, Chen Y<sup>#</sup>, Shu GP<sup>#</sup>, Zhao MM<sup>#</sup>, Zheng A, Chang XY, Li GQ, Wang YB<sup>\*</sup>, Zhang YM<sup>\*</sup>. Fast3VmrMLM: A fast algorithm that integrates genome-wide scanning with machine learning to accelerate gene mining and breeding by design for polygenic traits in large-scale GWAS datasets. *Plant Communications* 2025; 6: 101385.
- [2] Wang JT<sup>#</sup>, Han XL<sup>#</sup>, Zhao MM, Zhang HQ, Chen Y, Jiang QY and Zhang YM<sup>\*</sup>. A fast method for breeding by design via G×E interactions detected in large-scale climatic, phenomic and genomic data. *National Science Review* 2026, 13(10), nwag095.
- [3] Jing-Tian Wang<sup>#</sup>, Miao-Miao Zhao<sup>#</sup>, Xue-Lian Han<sup>#</sup>, Xiao-Yu Chang, Xiao-Shuang Ma, Qiu-Yun Jiang, Yuan-Ming Zhang<sup>\*</sup>. Fast3VmrMLM enables statistically powerful and computationally efficient large-scale marker-pair epistasis GWAS in sizable populations. *Plant Communications* 2026, in press.

### 4 Additional functions in Fast3VmrMLM for pre- and post-GWAS analyses

There were some functions that can be called by users in R environment for pre- and post-GWAS analyses in Fast3VmrMLM software.

#### 4.1 Draw a Manhattan plot

Users can draw a Manhattan plot by setting the parameter, *DrawPlot=TRUE*, when running the *Fast3VmrMLM()* and *Fast3VmrMLM\_MEJA()* functions. Alternatively, the flexible *Fast3VmrMLM\_manplot()* and *Fast3VmrMLM\_manplot\_MEJA()* function can be used to adjust the coordinate axis range and produce a lightweight PDF output. These functions can be accessed via the code below:

```
Fast3VmrMLM_manplot(midresult_dir="/home/trait1_intermediate.csv",
 result_dir="/home/trait1_result.xlsx",fileOut=NULL,Plotformat="*.tiff",c
```

*ol\_background=c("lightgreen","lightskyblue"),adj\_thres=150,log\_times=5,LOD\_times=3,reduction=0.5)*

***Fast3VmrMLM\_manplot\_MEJA***(*midresult\_dir="/home/trait1\_midresult.csv",result\_dir="/home/trait1\_result.xlsx",fileOut=NULL,Plotformat="\*.tif",col\_background=c("lightgreen","lightskyblue"),adj\_thres=150,log\_times=5,LOD\_times=3,reduction=0.5,loci\_type="QTN",chr\_axis\_size=0.5)*

| Parameter description: |                                                                                                                                                                                                    |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>midresult_dir</i>   | The intermediate result file output from GWAS (see § 2.2.1).                                                                                                                                       |
| <i>result_dir</i>      | The final result file output from GWAS (see § 2.2.1).                                                                                                                                              |
| <i>fileOut</i>         | The output address of Manhattan plot.                                                                                                                                                              |
| <i>Plotformat</i>      | The formate of output Manhattan plot. Supporting the formate of tiff, pdf, .png, and jpeg.                                                                                                         |
| <i>col_background</i>  | The color of the points.                                                                                                                                                                           |
| <i>adj_thres</i>       | The threshold for adjusting LOD score. When LOD score is larger than <i>adj_thres</i> the following adjustment was used to replace it: $LOD' = (LOD - adj\_thre) / 100 + adj\_thre$ .              |
| <i>log_times</i>       | The maximum value of the left y-axis is set to <i>log_times</i> times the largest log <i>P-value</i> .                                                                                             |
| <i>LOD_times</i>       | The maximum value of the right y-axis is set to <i>LOD_times</i> times the largest LOD score.                                                                                                      |
| <i>reduction</i>       | For large marker number ( <i>m</i> ), only a proportion ( <i>reduction</i> × <i>m</i> ) of markers is randomly sampled for plotting, which effectively reduces the size of the PDF Manhattan plot. |
| <i>loci_type</i>       | Only used in <b><i>Fast3VmrMLM_manplot_MEJA()</i></b> . Set <i>loci_type="QTN"</i> to plot QTN results or <i>loci_type="QEI"</i> to plot QEI results. The default is <i>loci_type="QTN"</i>        |

## 4.2 For individual-haplotype file

For the trait-associated haplotypes identified by Fast3VmrMLM-Hap, the individual-haplotype file records the haplotype of each individuals. It can be generated using *Fast3VmrMLM\_Hapout()* function via the following code:

**Fast3VmrMLM\_Hapout**(plinkFile="/home/gen",filePhe="/home/phe.csv",filePS=NULL,fileOut=NULL,file\_finalresult=NULL,c\_threshold=0.7,numofHaplotypes=3)

|                         |                                                                                                                                                               |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Parameter description:  |                                                                                                                                                               |
| <i>file_finalresult</i> | The final result file output from Fast3VmrMLM-Hap method (see § 2.2.1).                                                                                       |
| others                  | Other parameter should be same to the runing of Fast3VmrMLM-Hap.                                                                                              |
| Return value            | A file including all individuals and all haplotypes in the <i>file_finalresult</i> file. where "1" indicates the presence of the haplotype in the individual. |

### 4.3 For inferring the genotypes of offspring via pedigree and parents

The genotypes of offspring can be inferred from the pedigree and parental genotypes using the *Fast3VmrMLM\_outcode()* function via the following code. Before using this function, a dependency R packages, BEDMatrix should be installed in R environment via *install.packages("BEDMatrix")*.

The genotypes of offsprings can be inferred from the pedigree and parental genotypes using the *Fast3VmrMLM\_outcode()* function via the following code. Before using this function, the *BEDMatrix* R package dependency should be installed in the R environment via *install.packages("BEDMatrix")*.

**Fast3VmrMLM\_outcode**(plinkFile="/home/gen", filePedigree=NULL)

|                        |                                                                                                                                                                 |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Parameter description: |                                                                                                                                                                 |
| <i>plinkFile</i>       | The plink formate genotype file of parents.                                                                                                                     |
| <i>filePedigree</i>    | The pedigree file (see Table 7).                                                                                                                                |
| Return value           | A numerical matrix including all parents and offspring, where columns represent markers and rows represent individuals. Row names correspond to individual IDs. |

## 5 Other instructions and solutions of Issues

### 1) Detailed for installing Miniconda3

First, downdloading the installer of Miniconda3:

*wget https://repo.anaconda.com/miniconda/Miniconda3-py38\_23.11.0-2-Linux-x86\_64.sh*

Then, installing Miniconda3:

```
bash Miniconda3-py38_23.11.0-2-Linux-x86_64.sh
```

```
yes
```

```
yes
```

```
source ~/.bashrc
```

Third, confirm the installation successful:

```
conda --version
```

If displayed "conda 23.11.0", the installation was successful.

## **2) Issue**

There were "libstdc++.so.6: version 'CXXABI\_1.3.15' not found" reported after *library(Fast3VmrMLM)*.

**Solution:** Firstly, confirm that the R version is 4.3 and try to run Fast3VmrMLM again. If it doesn't work, update the version of libstdc++ by the following code:

```
cd /home/user/anaconda3/envs/Fast3VmrMLM/lib
```

```
rm libstdc++.so
```

```
rm libstdc++.so.6
```

```
ln -s libstdc++.so.6.0.34 libstdc++.so
```

```
ln -s libstdc++.so.6.0.34 libstdc++.so.6
```